CIVICS PROJECT

Analyzing and Reducing Water Consumption in AI Data Centers for Sustainable Climate Action



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Executive Summary

AI clusters consume millions of liters of freshwater daily for evaporative cooling. This paper evaluates three regulatory models against Indonesia's water laws and Pancasila social justice.

1. Problem Background

1.1 The Physical Footprint of Generative AI Factories

Generative artificial intelligence clusters operate as computational factories that diverge from standard cloud infrastructure. While conventional CPU workloads maintain dispersed power profiles, model training and inference require dense graphics processing unit (GPU) racks. Server racks housing Nvidia H100 and Blackwell B200 accelerators dissipate between 40 and 100 kilowatts of heat per rack. Because air cooling loses thermodynamic efficacy above 35 kilowatts per rack in tropical climates, operators rely on open evaporative cooling towers. Evaporative towers reject heat by spraying water across heat exchangers, venting steam into the atmosphere. A typical 100-megawatt facility evaporates 1.1 million to 2.5 million liters of freshwater daily. The Green Grid measures this through Water Usage Effectiveness (WUE), defined as annual site water consumption in liters divided by IT electricity consumption in kilowatt-hours. Standard evaporative facilities record a WUE between 1.5 and 2.2 L/kWh. Closed-loop dry chillers eliminate direct water evaporation, but raise electrical power draw by up to 15 percent under humid tropical weather conditions.

1.2 Indonesian Hydrological Realities and Tech Hub Expansion

Indonesia has become Southeast Asia's fastest-growing destination for hyperscale infrastructure. Following Singapore's 2019 data center moratorium caused by electrical and land limits, global cloud firms established facilities across West Java (Cikarang and Karawang) and Batam (Nongsa Digital Park). Government incentives through Special Economic Zone (KEK) policies accelerated this transition, yet facilities concentrate in hydrologically fragile corridors.

Java holds 56 percent of Indonesia's population but only 10 percent of total national water availability. The Citarum river basin and aquifers beneath Bekasi and Karawang suffer severe overdraft. Excessive industrial pumping causes land subsidence of 5 to 10 centimeters annually across northern coastal Java, salinizing municipal groundwater tables. Batam relies entirely on rain-fed surface reservoirs without natural river basins. Diverting millions of liters of municipal water to cool export AI compute while surrounding villages face water rationing generates severe resource conflict.

1.3 Civic Integrity, Anti-Corruption Culture, and Group 2 Mandate

This project is anchored in the Citizenship Education theme of Civic Integrity: A Student Movement to Build an Anti-Corruption Culture, aligned with Sustainable Development Goal 16 (Peace, Justice, and Strong Institutions) and Goal 6 (Clean Water and Sanitation). Unmonitored industrial water extraction breeds regulatory capture: when hyperscale facilities operate without public metering, opaque extraction licenses (SIPA) risk bureaucratic

collusion and rent-seeking. While Group 1 explains the empirical scale of this crisis, Group 2 focuses on reviewing alternative policies to safeguard public integrity, and provides the analytical basis for Group 3's forthcoming statutory proposals.

2. Data Collection Methods

2.1 Multi-Method Research Design

Because data center operators often treat facility water consumption as confidential operational data, relying on corporate reports creates substantial blind spots. We deploy a multi-method policy review structure combining technical secondary audits with qualitative stakeholder inquiry protocols to ensure rigorous statutory and empirical verification.

2.2 Statutory Documentation and Environmental Audits

Secondary data collection audits three regulatory streams: international standards (European Union Directive 2023/1791 and Singapore SS 564); Indonesian environmental statutes (Law No. 17 of 2019 on Water Resources, Government Regulation No. 22 of 2021 on Environmental Protection, and provincial Groundwater Extraction Permit / SIPA rules); and published Environmental Impact Assessments (AMDAL) from operators in Cikarang and Batam.

2.3 Field Interview Protocol and Guidelines

To evaluate operational compliance and regulatory friction, we established a structured qualitative interview protocol classifying informants into facility operations managers, environmental and river regulators, and municipal utilities/communities.

3. Analysis: Review of Alternative Policies

Governments deploy distinct regulatory instruments to manage water extraction from digital infrastructure. In accordance with Group 2's role, we review four primary policy alternatives: the European Union mandatory disclosure model, the Singapore and United States zoning moratorium model, the voluntary corporate offset model, and the existing Indonesian statutory baseline.

3.1 Policy Alternative 1: Mandatory Water Transparency and WUE Standards (The EU Benchmark)

This framework adapts the European Union Energy Efficiency Directive (Directive EU 2023/1791) and Germany's Energy Efficiency Act (EnEfG). Data centers exceeding 500 kilowatts must log operational metrics in a public database, detailing annual energy, total water withdrawal by source, and verified Water Usage Effectiveness (WUE). Public disclosure removes corporate secrecy, enabling regulators and citizens to track water

consumption, pushing operators toward direct-to-chip liquid cooling and closed-loop systems.

3.2 Policy Alternative 2: Spatial Zoning, Resource Moratoriums, and Industrial Quotas (The Singapore & US Benchmark)

This model relies on government zoning, drawn from Singapore's data center moratorium and Northern Virginia municipal ordinances. Authorities bar data center construction in critical groundwater basins outright. Outside prohibited zones, facilities face strict volumetric caps and cannot extract potable municipal supplies or groundwater, requiring non-potable sources like recycled wastewater or desalinated seawater.

3.3 Policy Alternative 3: Voluntary Corporate Environmental Stewardship and Water Offsets (The Silicon Valley Model)

This model reflects voluntary self-regulation favored by major tech firms, who announce corporate pledges such as becoming 'Water Positive by 2030.' Operators fund off-site watershed projects to match 100 to 120 percent of their water use. However, evaporating local groundwater today in Cikarang is not solved by funding reforestation hundreds of kilometers away, functioning largely as marketing tools that mask local water depletion.

3.4 Policy Alternative 4: The Existing Indonesian Statutory Baseline and Fiscal Distortions

Indonesia's current regulatory framework governing commercial water extraction is rooted in Law No. 17 of 2019 on Water Resources and Government Regulation (PP) No. 22 of 2021. Under this baseline, commercial entities must obtain a Groundwater Extraction Approval (Persetujuan Penggunaan Air Tanah) and operate under an Environmental Impact Assessment (AMDAL). However, operational enforcement suffers from institutional fragmentation, centralized verification deficits, and low extraction tariffs that distort sustainability incentives.

3.5 Theoretical and Constitutional Synthesis: Pancasila and Civic Integrity

Evaluating these policy choices requires connecting technical benchmarks to Article 33, Paragraph 3 of the 1945 Constitution (UUD 1945) and the Fifth Sila of Pancasila (Keadilan Sosial bagi Seluruh Rakyat Indonesia). In Decision No. 85/PUU-XI/2013, the Constitutional Court reaffirmed that water is an indivisible public right (res commune). When AI data centers extract millions of liters of potable groundwater daily while neighboring communities face dry wells, social justice is compromised. Enforcing transparent standards and non-potable mandates upholds constitutional rights and safeguards public integrity under SDG 16.

4. Conclusion

Our policy review shows that managing the water impact of AI data centers requires clear statutory standards. Voluntary corporate offsets (Alternative 3) fail to resolve acute local water scarcity, while the current statutory baseline (Alternative 4) remains weakened by low extraction fees and unverified monitoring. Neither model protects Indonesia's vulnerable water basins.

The analysis indicates that Indonesia must combine mandatory transparency standards (Alternative 1) with spatial zoning and non-potable water mandates (Alternative 2). Requiring public Water Usage Effectiveness (WUE) reporting and mandating closed-loop dry chillers in stressed corridors enforces Article 33 of the 1945 Constitution and the Fifth Sila of Pancasila. These findings provide the foundation for Group 3 to draft statutory reforms and for Group 4 to design the student civic action plan.

5. References

1. European Parliament and Council of the European Union. (2023). Directive (EU) 2023/1791 on energy efficiency and amending Regulation (EU) 2023/955 (recast). Official Journal of the European Union, L 231, 1–111.
2. Government of the Republic of Indonesia. (1945). The 1945 Constitution of the Republic of Indonesia (Undang-Undang Dasar Negara Republik Indonesia Tahun 1945). State Gazette of the Republic of Indonesia.
3. Government of the Republic of Indonesia. (2019). Law No. 17 of 2019 on Water Resources (Undang-Undang Republik Indonesia Nomor 17 Tahun 2019 tentang Sumber Daya Air). State Gazette of the Republic of Indonesia No. 190.
4. Government of the Republic of Indonesia. (2021). Government Regulation No. 22 of 2021 on the Implementation of Environmental Protection and Management. State Gazette of the Republic of Indonesia No. 32.
5. Indonesian Constitutional Court. (2015). Decision No. 85/PUU-XI/2013 on the Judicial Review of Law No. 7 of 2004 on Water Resources. Mahkamah Konstitusi Republik Indonesia.